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Identification modeling of ship maneuvering motion based on physics-informed neural network

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ABSTRACT

System identification is an effective method for modeling ship maneuvering motion, requiring only motion states and control inputs from model tests or full-scale trials. However, purely data-driven models may lack generalization due to the absence of physical constraints, and traditional parametric models can be restricted by simplified or approximate hydrodynamic formulations. To overcome this problem, a physics-informed neural network (PINN) is adopted. This method integrates the ship dynamic model into the loss function to improve robustness and enables the inverse identification of unknown hydrodynamic derivatives. A dynamic weighting strategy is introduced to enhance model performance, and sensitivity analysis is conducted. The method is validated on the KVLCC2 ship using simulation and SIMMAN experimental data. Results show that the PINN-based model can successfully predict ship maneuvering motion, and the prediction accuracy is satisfactory.

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System identification; modeling; ship maneuvering motion; physics-informed neural network; sensitivity analysis

1. Introduction

Driven by economic globalization, international trade has become more frequent, leading to a rise in the volume of goods transported. Ships, as one of the main modes of cargo transportation, are developing in the direction of large-scale and specialization. As a result, the complexity of ship maneuverability has increased. The International Maritime Organization (2002) has formulated standards for ship maneuverability, which proposes to predict ship maneuverability at the design stage. To this end, accurate ship maneuvering prediction has become a vital issue to be studied.

There are usually two methods for evaluating ship maneuverability at the initial design stage. One is to perform free-running tests, which can directly obtain the characteristic parameters including advance, transfer, tactical diameter, overshoot angles, etc. The other is to predict maneuvering performances based on system simulations. For the latter, simulation based on the mathematical model is one of the most effective methods for ship maneuverability prediction (Sutulo et al. 2002). An accurate ship dynamic model can not only predict ship maneuverability at the design stage but also help design the automatic control system of the ship (Sutulo and Guedes Soares 2014). Therefore, establishing a reliable ship dynamic model is of great practical value.

Common methods for ship dynamics modeling may include captive model tests, computational fluid dynamics (CFD), and system identification (SI). In the above methods, captive model tests and CFD calculation are reliable and effective modeling methods. However, captive model tests require specialized experimental equipment, and the CFD method requires more computational time. Moreover, captive model tests and the CFD method are difficult to apply to the full-scale ship. In contrast, the SI-based approach is an effective and convenient method that requires low experiment time and cost. This method only needs to obtain motion state information and control input collected during free-running model tests or full-scale trials, which makes it available for ships in both model and full scale.

There are two divisions for the SI-based modeling method: parametric and nonparametric modeling (Bai et al. 2019). Parametric modeling is based on the definite model structure. Many hydrodynamic

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derivatives in the model need to be determined. The commonly used SI methods for determining hydrodynamic derivatives may include ridge regression (Yoon and Rhee 2003), extended Kalman filter (EKF) (Perera et al. 2011), support vector machine (SVM) (Luo and Zou 2009; Zhang and Zou 2011; Wang et al. 2021), Bayesian optimization (BO) (Xue et al. 2020a), genetic algorithm (GA) (Sutulo and Guedes Soares 2014). The application of these SI methods has greatly advanced parametric modeling. However, issues such as parameter shift have always existed, which can potentially reduce the generalizability of the model (Luo and Li 2017). To diminish the parameter drift, researchers have proposed some methods (Hwang 1982; Yoon and Rhee 2003; Luo and Li 2017). Their results indicate that parameter drift cannot be eliminated. In addition, the model structure of parametric modeling is often based on the modeler's experience and knowledge. This limits the generalization performance of parametric modeling. To avoid the issues mentioned above, nonparametric modeling offers a potential approach.

The core concept of nonparametric modeling is that the model structure is assumed to be unknown, and the system is modeled based solely on the observed mapping relationship between the input and output datasets (Ouyang and Zou 2021). This method only requires defining the input and output forms of the training data, without being limited by any prior experience or knowledge. This makes it easy to implement and adaptable to any type of ship. Currently, many machine learning methods are available for nonparametric modeling of ship maneuvering motion. These methods can be broadly divided into two categories: kernel-based methods and neural networks. Kernel-based methods, such as SVM, Gaussian process regression (GPR), use kernel functions to model the nonlinear relationships in ship dynamics. Wang et al. (2015) applied least-squares support vector machine (LS-SVM) with radial basis kernel function to train a nonparametric model using simulation data from a 4-DOF ship. Then, ν-SVM (Wang et al. 2020), semblance-LS-SVM (Shen et al. 2023), twin SVM (Jiang et al. 2024) etc. have also been proposed to improve the modeling performance of SVM. Ariza Ramirez et al. (2018) were the first to use multi-output Gaussian process (GP) for nonparametric modeling of ship dynamics. Building on this, Xue et al. (2020b) proposed a GP method considering noise and conducted a comparative analysis using ship simulation data. Subsequently, Ouyang et al. (2023a, 2023b) made some improvements to GPR to improve the accuracy of nonparametric modeling. In addition, locally weighted learning (LWL) (Bai et al. 2019) and kernel ridge regression (KRR) (Moreno-Salinas et al. 2019) have also been applied to nonparametric modeling for ship maneuvering motion.

In contrast to kernel-based methods, neural networks rely on multiple layers of nonlinear transformations to capture the nonlinear characteristics. Their powerful nonlinear fitting capability offers an alternative for nonparametric modeling. Moreira and Guedes Soares (2003) applied recurrent neural networks (RNN) for nonparametric modeling of surface vessels, and their results indicated that neural networks have promising potential for practical application. Rajesh and Bhattacharyya (2008) proposed an artificial neural network (ANN) to model the nonlinear dynamics of large tankers. Zhang and Zou (2013) proposed a feedforward neural network with Chebyshev orthogonal basis functions. Their results showed that it outperforms traditional neural networks in nonparametric modeling of ship maneuvering motion. He et al. (2022) adopted a self-designed fully connected neural network to establish a nonparametric model for describing the ship motion and validated the method using model test data. With the development of deep learning, some deep neural networks have gradually been applied to nonparametric modeling of ship maneuvering motion (Woo et al. 2018; Lou et al. 2022; Zhang et al. 2024). From the above, it is evident that nonparametric modeling has significantly advanced ship maneuvering motion prediction. However, the accuracy and generalization performance of nonparametric modeling depend heavily on the quality and quantity of the training data. These models are also difficult to interpret, especially when faced with anomalies or unseen data. Furthermore, their data-driven nature makes them more sensitive to noise and prone to overfitting compared to parametric modeling (Wang et al. 2020).

Recently, Raissi et al. (2019) proposed a physics-informed neural network (PINN) and successfully applied it to solve nonlinear partial differential equations. This method improves the prediction accuracy and generalization performance by incorporating physical loss, derived from the physical equations, into the data-driven loss. Additionally, this method can also reverse-identify the unknown parameters in the physical equations. The idea of PINN provides a possible solution to overcome the drawbacks of nonparametric modeling. Xu et al. (2022) used PINN to predict the dynamics of unmanned surface vehicles. They constructed training samples with time as the input and velocity and angular velocity as the outputs. Their study directly employs a physics-based loss function, without separately considering the role of the data-driven loss. Zhao et al. (2024) adopted PINN to model the

dynamics of a 6-DOF autonomous underwater vehicle and conducted model predictive control research using the established PINN model. However, this study did not reverse-identify the hydrodynamic parameters but instead used known values. Guan et al. (2024) employed PINN to model the yaw motion of a deep-sea mining vehicle using simulation data. The weight of the physical loss in their PINN is determined empirically. Most of the above studies use high-quality training data and have not tested PINN on real measured data with significant noise. They also lack validation of its accuracy in identifying hydrodynamic derivatives. Furthermore, these studies primarily use loss functions with fixed weights and have not explored the effectiveness of dynamic weight adjustments.

In this paper, a PINN is employed for 3-DOF ship maneuvering motion modeling in calm water. To enhance the generalizability, the ship dynamic model is embedded as physical loss during training. The hydrodynamic derivatives in the dynamic model are unknown and need to be identified by PINN. A dynamic weight coefficient is assigned to the physical loss to accelerate PINN convergence and improve its modeling accuracy. To assess the effectiveness of PINN, two cases are implemented using simulated and experimental training data, respectively. A sensitivity analysis is conducted to evaluate the sensitivity of the hydrodynamic derivatives. The Savitzky–Golay filter is utilized to reduce noise in experimental data. The trained PINN model and identified hydrodynamic derivatives are both used for ship maneuvering prediction and a comparative analysis will be performed.

The rest of the paper is organized as follows: In Section 2, the ship mathematical model is described in detail. In Section 3, the modeling method based on PINN is presented. In Section 4, the case studies are carried out with data generated from simulation, and a sensitivity analysis is conducted. In Section 5, the experimental case studies are carried out with data from SIMMAN 2008. The experimental data are preprocessed first using the Savitzky–Golay filter, and the processed data are used as training data to validate the performance of the PINN-based model. Finally, Section 6 summarizes the conclusions.

2. Mathematical model of ship maneuvering motion

To describe the motion characteristics of a ship, as shown in Figure 1, two coordinate reference systems are adopted. One is the earth-fixed coordinates-$O_e - X_eY_eZ_e$. The other is the body-fixed coordinates $O_b - X_bY_bZ_b$.

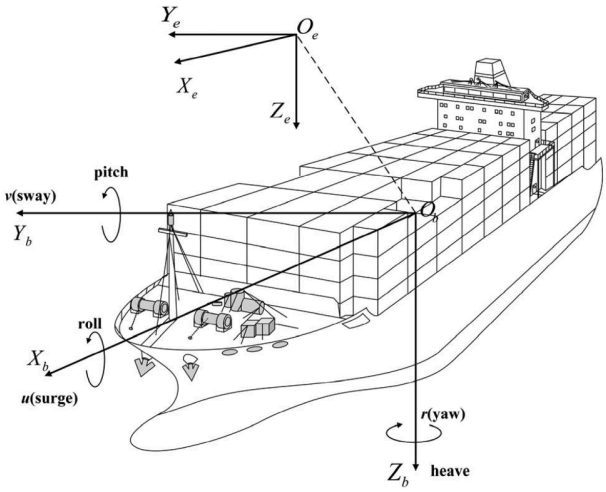


Figure 1. Coordinate reference systems of the ship. (This figure is available in color online.)

In this study, we only consider 3-DOF motion of the ship in deep water. According to Newton's law, the equations for surge motion, sway motion, and yaw motion can be written as follows:

$$\begin{aligned} m(\dot{u} - vr - x_G r^2) &= X \\ m(\dot{v} + ur + x_G \dot{r}) &= Y \\ I_Z \dot{r} + m x_G(\dot{v} + ur) &= N \end{aligned} \tag{1}$$

where u, v, $r$are the surge velocity, sway velocity and yaw rate, $m$is ship mass, I_Z is the moment of inertia about Z axis, x_G is the longitudinal coordinate of the ship gravity center in the body-fixed coordinate system, X, Y, Ndenote the longitudinal force, transverse force, and yaw moment. The dot over quantity expresses the operation of derivation concerning time.

Referring to the Abkowitz-type model (Abkowitz 1964), the right side of equation (1) may be approximated by following expressions:

$$\begin{aligned} X &= X_u \Delta u + X_{uu}(\Delta u)^2 + X_{vv}v^2 + X_{\dot{u}}\dot{u} + X_{rr}r^2 \\ &+ X_{vr}vr + X_{\delta\delta}\delta^2 + X_{v\delta}v\delta + X_{r\delta}r\delta \\ Y &= Y_v v + Y_{vvv}v^3 + Y_{\dot{v}}\dot{v} + Y_r r + Y_{rrr}r^3 + Y_{\dot{r}}\dot{r} \\ &+ Y_\delta \delta + Y_{\delta\delta}\delta^2 + Y_{\delta\delta\delta}\delta^3 + Y_{vrr}vr^2 + Y_{vvr}v^2 r + Y_{u\delta}\Delta u\delta \\ &+ Y_{v\delta\delta}v\delta^2 + Y_{vv\delta}v^2\delta + Y_{r\delta\delta}r\delta^2 + Y_{rr\delta}r^2\delta \\ N &= N_v v + N_{vvv}v^3 + N_{\dot{v}}\dot{v} + N_r r + N_{rrr}r^3 + N_{\dot{r}}\dot{r} + N_\delta \delta \\ &+ N_{\delta\delta}\delta^2 + N_{\delta\delta\delta}\delta^3 + N_{vrr}vr^2 + N_{vvr}v^2 r + N_{u\delta}\Delta u\delta \\ &+ N_{v\delta\delta}v\delta^2 + N_{vv\delta}v^2\delta + N_{r\delta\delta}r\delta^2 + N_{rr\delta}r^2\delta \end{aligned} \tag{2}$$

where $\Delta u = u - u_0$, and u_0 is the ship speed being taken as the expansion point of Taylor series, δ is the rudder angle, X_u etc. are the hydrodynamic derivatives.

Substituting equation (2) into equation (1). Then, let the acceleration terms be on the left side, and the other terms on the right side. The following equations are obtained:

$$\begin{aligned}(m - X_{\dot{u}})\dot{u} &= f_1 \\ (m - Y_{\dot{v}})\dot{v} + (mx_G - Y_{\dot{r}})\dot{r} &= f_2 \\ (mx_G - N_{\dot{v}})\dot{v} + (I_Z - N_{\dot{r}})\dot{r} &= f_3\end{aligned} \tag{3}$$

where f_1, f_2, f_3 are expressed as:

$$\begin{aligned}f_1 &= X_u\Delta u + X_{uu}(\Delta u)^2 + X_{vv}v^2 + (mx_G + X_{rr})r^2 \\ &+ (m + X_{vr})vr + X_{\delta\delta}\delta^2 + X_{v\delta}v\delta + X_{r\delta}r\delta \\ f_2 &= -mur + Y_v v + Y_{vvv}v^3 + Y_r r + Y_{rrr}r^3 + Y_\delta\delta \\ &+ Y_{\delta\delta}\delta^2 + Y_{\delta\delta\delta}\delta^3 + Y_{vrr}vr^2 + Y_{vvr}v^2r + Y_{u\delta}\Delta u\delta \\ &+ Y_{v\delta\delta}v\delta^2 + Y_{vv\delta}v^2\delta + Y_{r\delta\delta}r\delta^2 + Y_{rr\delta}r^2\delta \\ f_3 &= -mx_G ur + N_v v + N_{vvv}v^3 + N_r r + N_{rrr}r^3 \\ &+ N_\delta\delta + N_{\delta\delta}\delta^2 + N_{\delta\delta\delta}\delta^3 + N_{vrr}vr^2 + N_{vvr}v^2r \\ &+ N_{u\delta}\Delta u\delta + N_{v\delta\delta}v\delta^2 + N_{vv\delta}v^2\delta + N_{r\delta\delta}r\delta^2 + N_{rr\delta}r^2\delta\end{aligned} \tag{4}$$

By deriving the expressions for the acceleration terms, equation (3) can be rewritten as:

$$\begin{aligned}\dot{u} &= \frac{f_1}{m - X_{\dot{u}}} \\ \dot{v} &= \frac{(I_Z - N_{\dot{r}})f_2 - (mx_G - Y_{\dot{r}})f_3}{(m - Y_{\dot{v}})(I_Z - N_{\dot{r}}) - (mx_G - Y_{\dot{r}})(mx_G - N_{\dot{v}})} \\ \dot{r} &= \frac{(m - Y_{\dot{v}})f_3 - (mx_G - N_{\dot{v}})f_2}{(m - Y_{\dot{v}})(I_Z - N_{\dot{r}}) - (mx_G - Y_{\dot{r}})(mx_G - N_{\dot{v}})}\end{aligned} \tag{5}$$

$\dot{u}, \dot{v}, \dot{r}$can be obtained by discretization using the explicit Euler scheme as follows:

$$\begin{aligned}\dot{u} &= \frac{u(t+1) - u(t)}{\Delta t}, \ \dot{v} = \frac{v(t+1) - v(t)}{\Delta t}, \\ \dot{r} &= \frac{r(t+1) - r(t)}{\Delta t}\end{aligned} \tag{6}$$

where Δt is the time step, $t$and$t+1$are two consecutive time steps.

From the derivation above, once the current acceleration is obtained, the next motion state can be iteratively calculated using equation (6). For this end, a physics-informed neural network is employed to predict the acceleration value at the next moment, as described below.

3. Modeling method based on PINN

3.1. Neural network

In this study, a feedforward neural network is employed for ship maneuvering motion modeling, with a structure consisting of an input layer, multiple hidden layers, and an output layer. The neurons between two adjacent layers are connected in pairs, each with an associated weight, and each neuron also has a bias. The process of passing input data through the neural network to the output can be described as follows:

$$\begin{cases} X_{in} = a^{(0)} \\ z^{(l)} = W^{(l)}a^{(l-1)} + b^{(l)}, l = 1, 2, \cdots L \\ a^{(l)} = f_a(z^{(l)}) \\ Y_{out} = f_a(W^{(L)}a^{(L-1)} + b^{(L)}) \end{cases} \tag{7}$$

where X_{in} is the input vector, Y_{out} is the predicted output, L represents the number of layers, $a^{(l)}$ is the output of the l-th layer, $W^{(l)}$ and $b^{(l)}$ are the weight matrix and bias vector of l-th layer respectively, both of which are trainable parameters in the network, $f_a(\cdot)$ is the nonlinear activation function, which enables neural network to deal with nonlinear problem. In this study, the Tanh activation function is used.

To train neural networks, a loss function is constructed to measure the error. The gradients of the loss function with respect to the network parameters are computed using the chain rule. Finally, the network parameters are updated using Adam algorithm (Kingma and Ba 2014). To enhance convergence stability and prevent overfitting, the following learning rate decay strategy is incorporated into the Adam optimizer:

$$lr = 0.001 \cdot (0.1)^{\left[\frac{iter}{40000}\right]} \tag{8}$$

where lr denotes learning rate, $iter$ is the number of training iterations $[\cdot]$ denotes a downward rounding operation.

3.2. Physics-informed neural network

The physics-informed neural network (PINN), proposed by Raissi et al. (2019), is a deep learning method

used to solve partial differential equations. The core idea of PINN is to implicitly embed the physical knowledge described by the partial differential equations into the neural networks as a loss function. Moreover, this method can also be used to inverse-solve unknown parameters in the physical equations. The specific algorithm is introduced as follows:

Consider a problem that can be described by the following partial differential equation:

$$\begin{cases} \partial_t + D[\partial, \lambda] = 0, x \in \Omega, t \in (0, T) \\ \partial(x, t) = I(x, t) \end{cases} \tag{9}$$

Where $\partial(x, t)$denotes the latent solution of the partial differential equation, $D[\cdot, \lambda]$ is a nonlinear differential operator, λ are unknown parameters to be determined, $x$is the spatial variable, $t$is the time variable, Ω is the spatial domain, $I(x, t)$ is the initial or boundary conditions.

Assume a neural network with parametersθ, where the inputs to the network are x and t, the output is $\partial(x, t; \theta)$. To integrate the above physical equation into the neural network, the loss function is constructed as follows.

$$f := \partial_t + D[\partial, \lambda] \tag{10}$$

$$Loss_F = \frac{1}{N_F}\sum_{i=1}^{N_F} |f(x_F^i, t_F^i; \theta)|^2 \tag{11}$$

$$Loss_I = \frac{1}{N_I}\sum_{i=1}^{N_I} |\partial(x_I^i, t_I^i; \theta) - I(x_I^i, t_I^i)|^2 \tag{12}$$

$$Loss_D = \frac{1}{N_D}\sum_{i=1}^{N_D} |\partial(x_D^i, t_D^i; \theta) - \partial_D^i|^2 \tag{13}$$

where $\{x_F^i, t_F^i\}_{i=1}^{N_F}$ denote sampling points in the time and spatial domains, $f$is the partial differential equations residual, $\{x_I^i, t_I^i, I(x_I^i, t_I^i)\}_{i=1}^{N_I}$ represent the initial or boundary training data on$\partial(x, t)$, $\{x_D^i, t_D^i, \partial_D^i\}_{i=1}^{N_D}$ represents observation data. Finally, the total loss function can be expressed as:

$$Loss = Loss_F + Loss_I + Loss_D \tag{14}$$

3.3. Ship maneuvering motion modeling based on PINN

Following the idea of PINN, the ship dynamic equations need to be modified to the form of equation (9). To this end, set the inputs to the neural network as velocity terms u, v, r with rudder angle δ and the outputs as accelerations $\dot{u}$, $\dot{v}$, $\dot{r}$. The initial and boundary conditions are not taken into account here. According to equation (5), the constraint equation can be constructed as:

$$e_1 = \left| \frac{f_1}{m - X_{\dot{u}}} - \dot{u} \right|$$

$$e_2 = \left| \frac{(I_Z - N_{\dot{r}})f_2 - (mx_G - Y_{\dot{r}})f_3}{(m - Y_{\dot{v}})(I_Z - N_{\dot{r}}) - (mx_G - Y_{\dot{r}})(mx_G - N_{\dot{v}})} - \dot{v} \right|$$

$$e_3 = \left| \frac{(m - Y_{\dot{v}})f_3 - (mx_G - N_{\dot{v}})f_2}{(m - Y_{\dot{v}})(I_Z - N_{\dot{r}}) - (mx_G - Y_{\dot{r}})(mx_G - N_{\dot{v}})} - \dot{r} \right| \tag{15}$$

where $\dot{u}$, $\dot{v}$, $\dot{r}$are predicted by the neural network, f_1, f_2, f_3 are calculated from equation (4). The hydrodynamic derivatives in equation (4) are unknown parameters and are treated as independent trainable variables in the neural network. These derivatives are incorporated into the physics-based constraints and are jointly optimized with the neural weights using the Adam algorithm during the minimization of the overall loss function.

According to the equation (15), the physical loss function is formulated as:

$$Loss_{phy} = \frac{1}{N_o}\sum_{i=1}^{N_o} (e_1^2 + e_2^2 + e_3^2) \tag{16}$$

where $N_o$denotes the number of training data. The loss function for observation data is as follows:

$$Loss_{data} = \frac{1}{N_o}\sum_{i=1}^{N_o} (|\dot{u}_{data} - \dot{u}|^2 + |\dot{v}_{data} - \dot{v}|^2 + |\dot{r}_{data} - \dot{r}|^2) \tag{17}$$

where $\dot{u}_{data}$, $\dot{v}_{data}$, $\dot{r}_{data}$ denote the observed acceleration values. The total loss function is as follows:

$$Loss = Loss_{data} + \beta \cdot Loss_{phy} \tag{18}$$

where β is the dynamic weight coefficient, with an initial value of 0.01. After 10,000 iterations, it becomes 0.1, and after 50,000 iterations, it reaches 1. The reason for this approach is that the hydrodynamic derivatives in the physical equations are unknown. Therefore, in the early stages of training, the neural network focuses more on fitting the data to learn the underlying patterns. As training progresses, the weights of the physical loss and data-driven loss are gradually balanced. This process guides the network to satisfy the physical laws while still maintaining good data fitting. The whole modeling process of ship maneuvering motion based on PINN is shown in Figure 2.

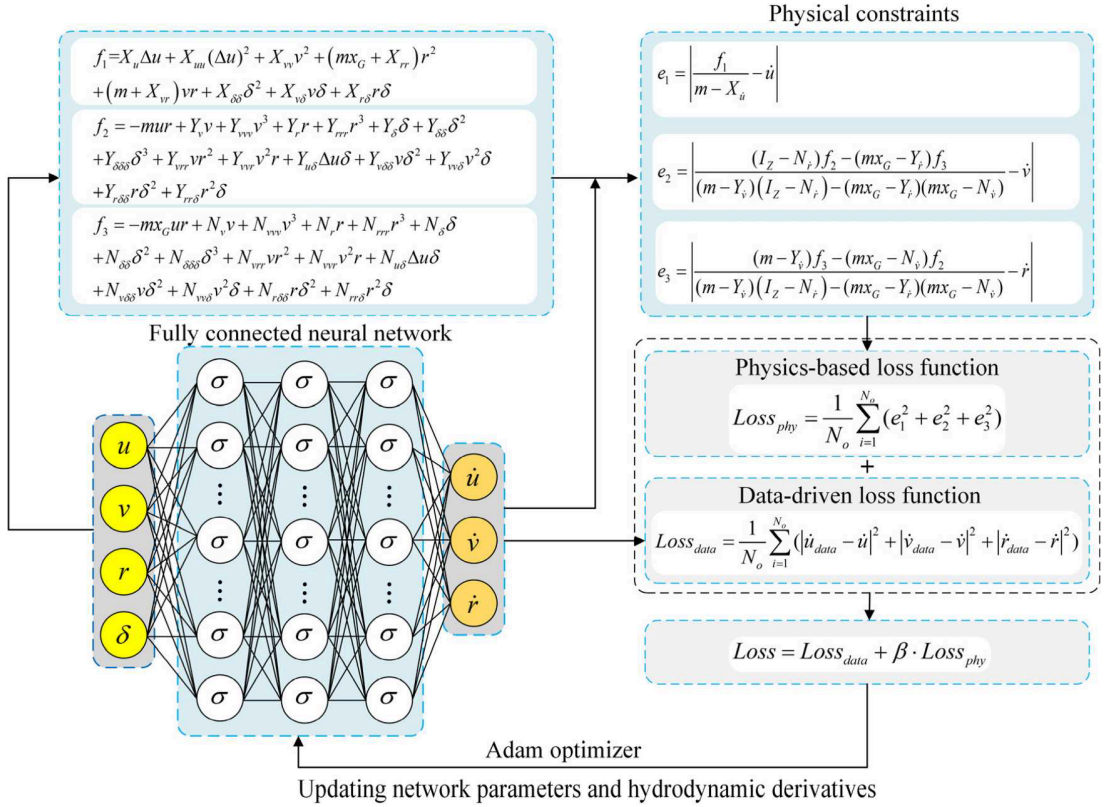


Figure 2. Flowchart of ship maneuvering motion modeling based on PINN. (This figure is available in color online.)

4. Case study with simulation data

4.1. Simulation data acquisition

The second variant of Korea Very Large Crude Carrier (KVLCC2) is selected for the present case study. It is a widely used benchmark ship for validation of maneuvering prediction methods. The main particulars of KVLCC2 are shown in Table 1.

To evaluate the effectiveness of PINN, the ship maneuvering motion modeling is carried out using simulation data. To this end, the hydrodynamic derivatives from literature (Yao et al. 2021) are used to solve equations (4)–(6). This process generates the simulation training data, which includes 35° turning circle maneuver, 10°/10° and 20°/20° zigzag maneuvers. The sampling interval is 1s, with each motion sampled for 1000s, resulting in a total of 3000 data points used for PINN training.

To prevent large-scale features from dominating smaller ones, which could lead to algorithm instability or reduced accuracy, data standardization is applied before training (Wang et al. 2019). Each feature variable x is standardized using the following formula:

$$x\prime = \frac{x-\mu}{\sigma} \tag{19}$$

where $x\prime$ is the standardized variable, μ is the mean value of the variable, σ is the variance of the variable.

Table 1. Main particulars of KVLCC2.

Parameters	Values
Length between perpendiculars (L_{pp})	320.00 m
Breadth	58.00 m
Design draft	20.80 m
Block coefficient	0.81
x coordinate of CG (x_G)	11.136 m
Inertia in yaw (I_Z)	2.00 × 10^{12} kg/m^2
Rudder turning rate	2.34°/s
Approach speed	7.956 m/s

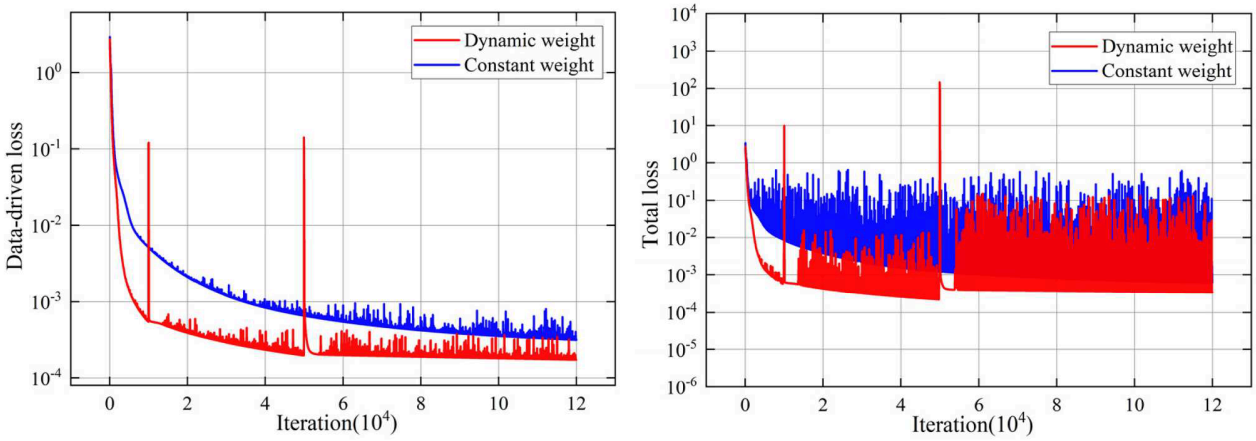
4.2. Identification with simulation data

According to study (He and Zou 2020), shallow networks are adequately effective for modeling the 3-DOF ship maneuvering motion in calm water. Therefore, based on this experience, a PINN with three hidden layers, each containing 10 neurons, is applied in this study. The Xavier (Glorot and Bengio 2010) method is used to initialize the weights and biases in the neural network. The initial values of the hydrodynamic derivatives are set to 0. The network parameters and hydrodynamic derivatives are optimized iteratively using the Adam algorithm. Since the acceleration derivatives can

be computed using a mature method, it is not considered in the identification process. The values of the acceleration derivatives are as follows: $X_{\dot{u}}\prime = -0.00114$, $Y_{\dot{v}}\prime = -0.0145$, $Y_{\dot{r}}\prime = -0.00121$, $N_{\dot{v}}\prime = -0.000588$, and $N_{\dot{r}}\prime = -0.000564$. All studies are carried out in a uniform environment running PyCharm 2023 version with an Intel(R) Core (TM) i7-12,700 K, 3.60 GHz CPU. The PINN is trained for 120,000 iterations. The training time is approximately 8 min. To validate the effectiveness of the dynamic weight β in equation (18), a total loss function with a constant weight of 1 is also used for comparison. The comparison results of the convergence processes between the constant weight and the dynamic weight are shown in Figure 3. The identification results of hydrodynamic derivatives using PINN are shown in Table 2.

As shown in Figure 3(a), during the early training stages, the data-driven loss with dynamic weight is significantly lower than that with constant weight. This is because, at this stage, the lower physical weight enables the neural network to quickly learn the features associated with data. In contrast, the data-driven loss with constant weight will be affected by the uncertain hydrodynamic derivatives in the physical constraints. Although the physical weight increases with iterations in the dynamic weight strategy, the data-driven loss remains lower than that with constant weight. This shows that the dynamic weight strategy maintains good data fitting ability while satisfying physical rules.

As shown in Figure 3(b), during the early training stages, the total loss with dynamic weight decreases rapidly. This is attributed to the data-driven loss playing a dominant role in the total loss. As the weight of physical loss dynamically increases, the total loss shows two sharp increases. Subsequently, the weights of physical and data-driven losses reach a balance. At the end of training, the total loss with dynamic weight shows a generally lower value than that with



(a) Convergence process of data-driven loss (b) Convergence process of total loss

Figure 3. Comparison of convergence process based on simulation data. (This figure is available in color online.)

Table 2. Comparison of dimensionless hydrodynamic derivatives using PINN based on simulation data.

X-Coef.	Original	Identified	Y-Coef.	Original	Identified	N-Coef.	Original	Identified
X'_u	−0.00319	−0.0032	Y'_v	−0.01902	−0.0193	N'_v	−0.007886	−0.0078
X'_{uu}	0.00199	0.0020	Y'_{vvv}	−0.1287	−0.1024	N'_{vvv}	0.00175	0.0102
X'_{vv}	0.00159	−0.0028	Y'_r	0.005719	0.0056	N'_r	−0.003701	−0.0037
X'_{rr}	0.000338	−0.0009	Y'_{rrr}	−0.000048	0.000045	N'_{rrr}	−0.000707	−0.0011
X'_{vr}	0.01391	0.0092	Y'_{vrr}	−0.02429	−0.0167	N'_{vrr}	0.003726	0.0054
$X'_{\delta\delta}$	−0.00272	−0.0027	Y'_{vvr}	0.0211	0.0493	N'_{vvr}	−0.019	−0.0100
$X'_{v\delta}$	0.001609	0.0008	Y'_{δ}	0.00408	0.0040	N'_{δ}	−0.001834	−0.0018
$X'_{r\delta}$	−0.001034	−0.0015	$Y'_{\delta\delta}$	−0.000114	−0.0001	$N'_{\delta\delta}$	−0.000056	−0.000057
			$Y'_{\delta\delta\delta}$	−0.003059	−0.0029	$N'_{\delta\delta\delta}$	0.001426	0.0014
			$Y'_{u\delta}$	−0.00456	−0.0046	$N'_{u\delta}$	0.00232	0.0023
			$Y'_{v\delta\delta}$	0.00326	0.0050	$N'_{v\delta\delta}$	−0.001504	−0.0014
			$Y'_{vv\delta}$	0.003018	0.0053	$N'_{vv\delta}$	−0.001406	0.0001
			$Y'_{r\delta\delta}$	−0.002597	−0.0017	$N'_{r\delta\delta}$	0.001191	0.0011
			$Y'_{rr\delta}$	0.000895	0.0003	$N'_{rr\delta}$	−0.000398	−0.0010

constant weight. This indicates that the dynamic weight strategy provides a more effective way to incorporate physical constraints.

Table 2 shows that most of the identified hydrodynamic derivatives are close to the original values, while some higher-order hydrodynamic derivatives exhibit significant deviations. This discrepancy is likely due to the optimization process not converging to a global optimum. Furthermore, sensitivity analysis of the hydrodynamic derivatives indicates that some higher-order derivatives have a small contribution to the maneuvering motion (Wang et al. 2014). This makes their values prone to large fluctuations during the identification process. To evaluate the sensitivity of the hydrodynamic derivatives used in this study, a sensitivity analysis is conducted as follows.

4.3. Sensitivity analysis

Elements of the sensitivity matrix can be approximated by

$$S_j^i = \frac{\partial \xi_j}{\partial \alpha_i} = \frac{\xi_j(\alpha_i + \Delta\alpha_i) - \xi_j(\alpha_i)}{\Delta\alpha_i} \qquad j = 1, 2, 3; \quad i = 1, 2, \cdots N_c \tag{20}$$

where $\xi = [u, v, r]^T$is the state vector of ship motion, and ξ_j denotes the j-th component of ξ, S_j^i denotes the sensitivity value of the hydrodynamic derivative-$\alpha_i$with respect to the motion parameter ξ_j, N_c is the number of hydrodynamic derivatives. To determine the sensitivity of hydrodynamic derivatives, the direct method (Yeo and Rhee 2006) is employed for this study. The calculation process of the direct method is as follows:

The mathematical model of ship maneuvering motion can be expressed in the form of the following differential equations:

$$M(\alpha)\frac{d\xi}{dt} = f(\xi, \alpha) \tag{21}$$

where $M(\alpha)$ represents the inertia matrix of the ship, $f = [f_1, f_2, f_3]^T$ are the nonlinear functions related to the state information.

Multiplying equation (21) with the inverse of the inertia matrix yields equation (22)

$$\frac{d\xi}{dt} = M^{-1}(\alpha) \cdot f(\xi, \alpha) = g(\xi, \alpha) \tag{22}$$

According to equations (20) and (22), the sensitivity differential equation with respect to the hydrodynamic derivatives can be obtained as follows:

$$\dot{S} = \frac{d}{dt}\frac{\partial \xi}{\partial \alpha} = \frac{\partial g(\xi, \alpha)}{\partial \xi} \cdot \frac{\partial \xi}{\partial \alpha} + \frac{\partial g(\xi, \alpha)}{\partial \alpha} = J \cdot S + G \tag{23}$$

where S is the sensitivity matrix, JandGare the system Jacobian and hydrodynamic derivatives Jacobian, respectively, which can be expressed as:

$$J = \frac{\partial g(\xi, \alpha)}{\partial \xi} = M^{-1}(\alpha) \cdot \frac{\partial f(\xi, \alpha)}{\partial \xi}$$

$$G = \frac{\partial g(\xi, \alpha)}{\partial \alpha} = \frac{\partial M^{-1}(\alpha)}{\partial \alpha} \cdot f(\xi, \alpha) + M^{-1}(\alpha) \cdot \frac{\partial f(\xi, \alpha)}{\partial \alpha} \tag{24}$$

Since the acceleration derivatives are already known in this study, they are not included in the sensitivity analysis. Equation (23) is simplified as follows:

$$\frac{dS}{dt} = M^{-1}(\alpha) \cdot \left[\frac{\partial f(\xi, \alpha)}{\partial \xi} \cdot S + \frac{\partial f(\xi, \alpha)}{\partial \alpha}\right] \tag{25}$$

The motion state variables can be obtained by performing ship maneuvering motion simulation. Then, the system Jacobian J and the hydrodynamic derivatives Jacobian G can be calculated using the motion state variables. Finally, the sensitivity matrix S can be determined by numerically solving equation (25).

The sensitivity value calculated from equation (25) is dimensional, which is not convenient for comparison. For this, normalization of the sensitivity value is applied. The normalized sensitivity matrix can be defined as

$$S_j^i = \frac{\alpha_i}{\xi_{j,std}}\frac{\partial \xi_j}{\partial \alpha_i} \tag{26}$$

where $\xi_{j,std} = [1\ \text{m/s},\ 1\ \text{m/s},\ 0.01\,\text{rad/s}]^T$.

According to the above calculation, the time histories of the sensitivity values in ship maneuvering motion simulation can be obtained. It represents the trend of the sensitivity of each hydrodynamic derivative over time, but it cannot directly analyze the importance of each derivative to the maneuvering motion. To quantitatively compare the importance of each derivative, the sensitivity value of the hydrodynamic derivative with respect to each motion state ($\tilde{S}_j^i$) is defined as

$$\tilde{S}_j^i = \frac{\sum_{h=1}^{N_s} |S_j^i(h)|}{\sum_{i=1}^{N_c}\sum_{h=1}^{N_s} |S_j^i(h)|} \times 100\% \tag{27}$$

where h is the sampling time step, N_s is the total number of the sampling time steps.

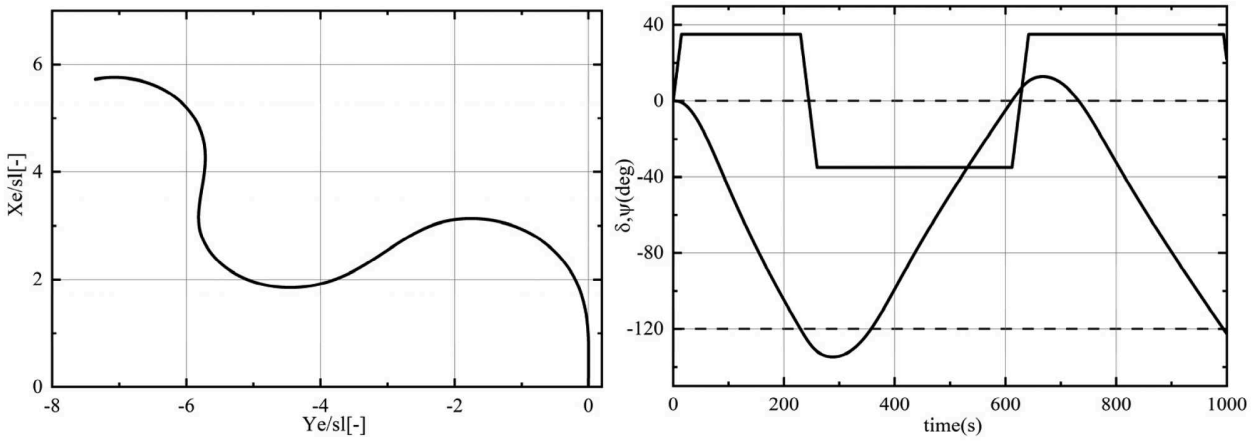


Figure 4. The simulation results of 120° S-type test for sensitivity analysis. (This figure is available in color online.)

In Rhee and Kim's work (Rhee and Kim 1999), the results show that zigzag motion and turning circle motion are not suitable for sensitivity analysis. They proposed a new test named as S-type test. In this study, a 120° S-type test is used for sensitivity analysis. The trajectory of a 120° S-type test, and the corresponding rudder angle and heading angle are shown in Figure 4. The sensitivity analysis results are shown in Figure 5.

As shown in Figure 5, X'_{vv}, X'_{rr}, $X'_{v\delta}$ and $X'_{r\delta}$ have a relatively minor influence on surge motion; Y'_{rrr}, $Y'_{\delta\delta}$, $Y'_{v\delta\delta}$, $Y'_{vv\delta}$, $Y'_{r\delta\delta}$ and $Y'_{rr\delta}$have a limited influence on sway motion; N'_{vvv}, N'_{rrr}, $N'_{\delta\delta}$, $N'_{v\delta\delta}$, $N'_{vv\delta}$, $N'_{r\delta\delta}$ and $N'_{rr\delta}$ have a slight influence on yaw motion. Comparison with Table 2 shows that hydrodynamic derivatives with lower sensitivity exhibit larger identification errors, while those with higher sensitivity show smaller errors. This suggests that hydrodynamic derivatives with low sensitivity are more prone to parameter drift. At the same time, it also indicates that the identification errors of these hydrodynamic derivatives do not significantly affect the prediction performance.

4.4. Generalization test with simulation data

To assess the generalization performance of the PINN-based model, two prediction approaches are employed: one directly predicting maneuvering motion with the PINN model, and the other using the identified hydrodynamic derivatives from the PINN to predict maneuvering motion through the Abkowitz-type model. Both approaches are used to predict the maneuvers not involved in the training data, including 25°/5° zigzag maneuver, 20°/10° zigzag maneuver, and 30° turning circle maneuver. The comparison between the predicted results and the simulation data is shown in Figures 6–8. Additionally, the root mean square error (RMSE) is used to quantify prediction accuracy, with the formula given as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2} \tag{28}$$

where $\hat{y}_i$ is the predicted value, and y_i is the original data, $n$represents the number of predicted samples. The RMSEs of predicted results are shown in Table 3.

Figures 6–8 show that the predicted results of identified model and PINN model match well with the original simulation data. This indicates that the PINN-based model exhibits good generalization performance when trained with simulation data. As shown in Table 3, the RMSE values indicate no significant difference in prediction accuracy between the two models. This may be attributed to the fact that, in the later stages of PINN training, the balance between the physical loss and data-driven loss results in similar predictive performance. However, it should be noted that simulation data is noise-free; the performance of the PINN-based model with real measured data still needs further validation. Overall, the predictions based on simulation data are relatively accurate.

5. Case study with experimental data

5.1. Experimental data acquisition

In order to further verify the effectiveness of PINN, the ship maneuvering motion modeling is carried out using the real measured data from the SIMMAN 2008 workshop. The free-running model tests were performed using the KVLCC2 ship model at Maritime Research Institute Netherlands (MARIN). The main

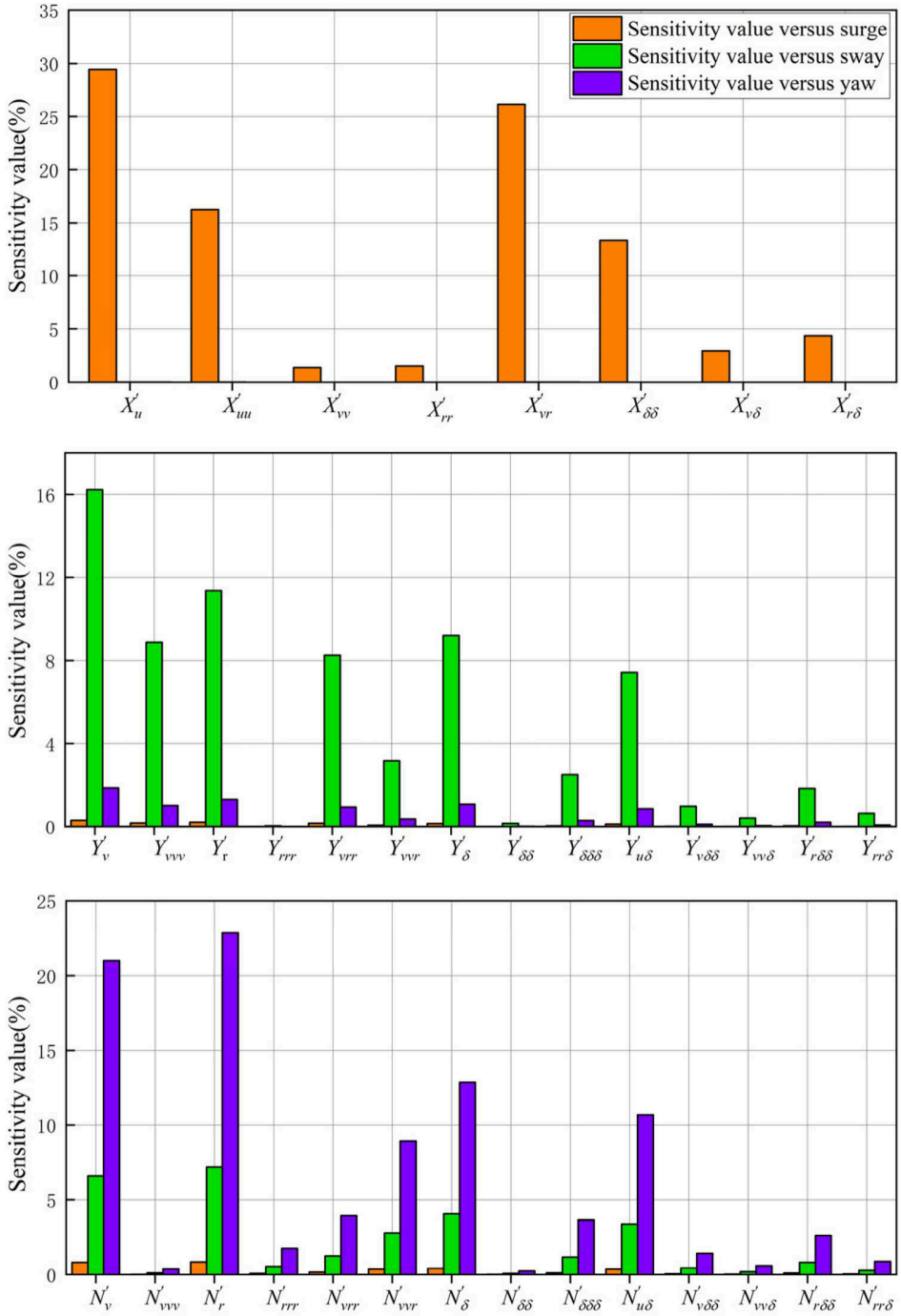


Figure 5. Results of sensitivity analysis. (This figure is available in color online.)

Table 3. Estimation of prediction accuracy by RMSE based on simulation data.

		u	*v*	*r*	sum
25°/5° zigzag	PINN	0.0398	0.0909	0.0419	0.1726
	Identified	0.0471	0.0669	0.0327	0.1467
20°/10° zigzag	PINN	0.0130	0.0145	0.0063	0.0338
	Identified	0.0281	0.0186	0.0083	0.0550
30° turning circle	PINN	0.0508	0.0041	0.0013	0.0562
	Identified	0.0376	0.0070	0.0025	0.0471

particulars of KVLCC2 are shown in Table 1. The dataset includes six maneuvering tests: 10°/10° zigzag test (P), 10°/10° zigzag test (S), 20°/20° zigzag test (P), 20°/20° zigzag test (S), 35° turning circle test (P), and 35° turning circle test (S). 'P' indicates a maneuver starting from the port side, and 'S' indicates a maneuver starting from the starboard. All the data are presented in full scale.

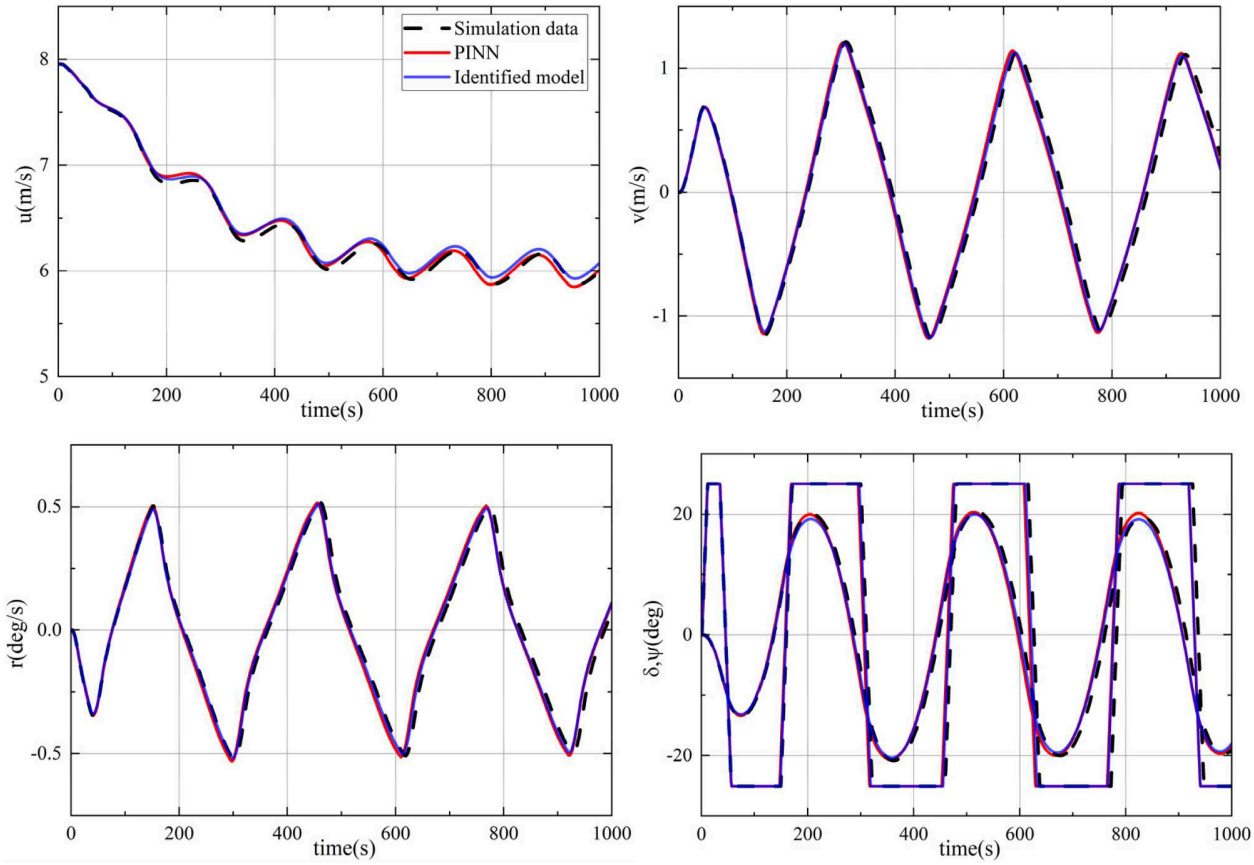


Figure 6. Prediction of 25°/5° zigzag maneuver based on simulation data. (This figure is available in color online.)

As is well known, experimental data often contains significant noise due to sensor errors and various external disturbances, such as wind, waves, and currents. The presence of noise increases the uncertainty in PINN modeling and may lead to overfitting. Moreover, these disturbances can sometimes have a greater impact on the ship than the partial hydrodynamic derivatives. This makes accurate identification of the hydrodynamic derivatives more challenging. Therefore, it is essential to minimize the impact of noise. For this purpose, experimental data is preprocessed using the Savitzky–Golay filter to reduce noise in this study.

5.2. Savitzky–Golay filter

The Savitzky–Golay filter, proposed by Savitzky and Golay (1964), smooths data based on polynomial fitting within a moving window. Its simple parameters, high computational efficiency, and ability to preserve key signal features make it widely used in data denoising. The basic principle is as follows:

Assume that there are $2M + 1$ points within a moving window centered at $i = 0$, and the data within this window is fitted using the following polynomial:

$$p(i) = \sum_{k=0}^{K} d_k i^k,\ K \le 2M + 1 \tag{29}$$

where K is the degree of the polynomial, d_k are the coefficients of the polynomial, $i$is the discrete sequence value at this point, $p(i)$ is the fitted value at this point, $M$is the half-width of the moving window.

Next, the residuals between all the fitted values and the original data within the moving window are constructed as follows:

$$\varepsilon = \sum_{i=-M}^{M} (p(i) - s(i))^2 = \sum_{i=-M}^{M} (\sum_{k=0}^{K} d_k i^k - s(i))^2 \tag{30}$$

where $s(i)$ represents the original data point. A smaller value of εindicates a better fit. To this end, the least squares criterion expressed by equation (31) is used to minimize $\varepsilon$and solve for the polynomial coefficients. Then, the smoothed result is obtained by inputting the window's center point coordinates into the derived polynomial. Finally, the window is moved, and the above

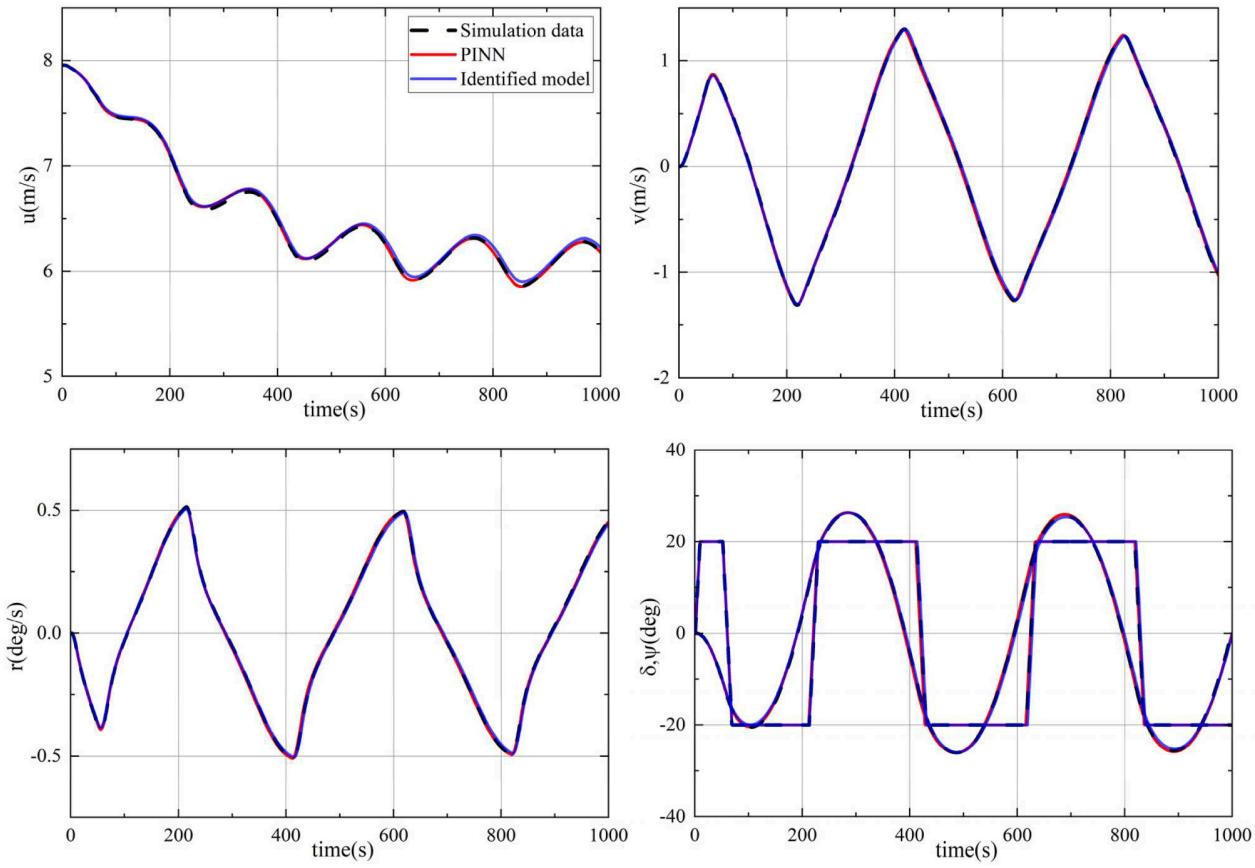


Figure 7. Prediction of 20°/10° zigzag maneuver based on simulation data. (This figure is available in color online.)

steps are repeated until the entire dataset is denoised.

$$\frac{\partial}{\partial d_k}\left[\sum_{i=-M}^{M}\left(\sum_{k=0}^{K} d_k i^k - s(i)\right)^2\right] = 0 \tag{31}$$

As seen from the above process, once the polynomial degree and window size are determined, the Savitzky–Golay filter can quickly smooth the data. In this study, the polynomial degree K is set to 3, and the half-width-$M$of the window is set to 100. Following the above approach, the Savitzky–Golay filter is first applied to smooth the velocity data. The smoothed velocity data is used to calculate the acceleration, as given by equation (6), which is then smoothed again using the Savitzky–Golay filter. As an example, the smoothed results of the velocity values$u$, v and acceleration values $\dot{u}$, $\dot{v}$from the 20°/20° zigzag test (S) are shown in Figures 9 and 10.

5.3. Validation on experimental data

In this section, the measured data from SIMMAN 2008 will be divided into two sets: the training set and the test set. The data in the training set will be preprocessed with the Savitzky–Golay filter. Meanwhile, data standardization of the training set is performed using equation (19). The training set, consisting of all data from10°/10° zigzag test (P) and 20°/20° zigzag test (S), and the first 10,000 data points from the 35° turning circle test (which includes both P and S test data). The sampling interval is 0.135s, and the training set contains a total of 28,896 sample points. The test set, made up of 10°/10° zigzag test (S), 20°/20°zigzag test (P), and 35° turning circle test (P), will be used to evaluate the model's accuracy and generalization performance. The parameter settings of PINN are the same as those in Section 4.2. The initial values of the hydrodynamic derivatives are still set to 0. The comparison results of the convergence processes between the constant weight and the dynamic weight are shown in Figure 11. The identification results using PINN based on experimental data are shown in Table 4. In addition to the PINN, a neural network (NN) without embedded physical constraints is used for comparison. This model shares the same parameter settings and training process as the PINN. The training times for PINN and NN are approximately 16 and 5 min, respectively. While the PINN requires a longer training time than the NN,

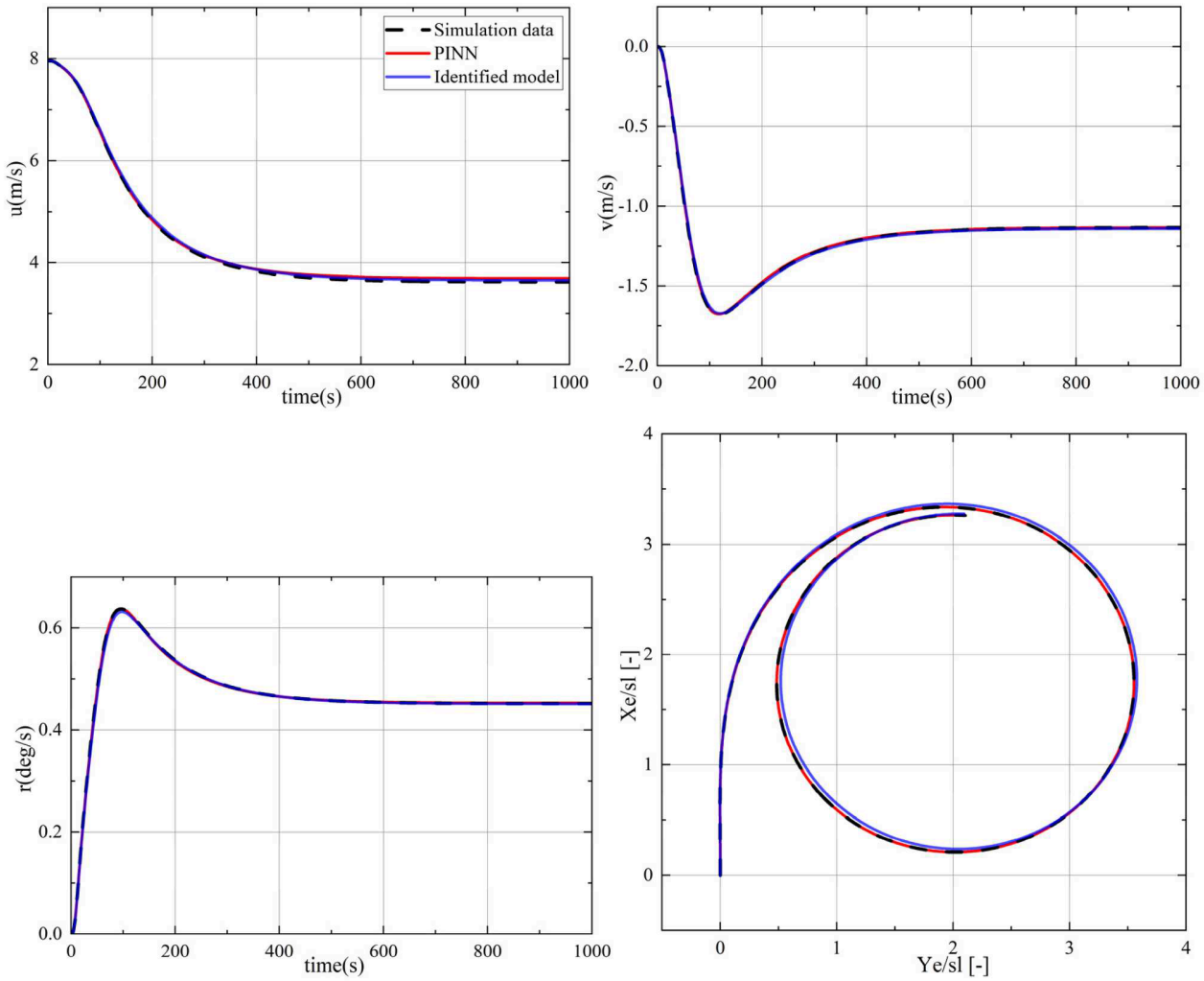


Figure 8. Prediction of 30° turning circle maneuver based on simulation data. (This figure is available in color online.)

incorporating physical constraints may enhance generalization performance and enable identification of hydrodynamic derivatives, at only a moderate increase in computational cost. The predicted results for all models are shown in Figures 12–14. The RMSEs of predicted results for all models are shown in Table 5.

As shown in Figure 11(a), the data-driven loss with dynamic weight is lower than with constant weight. In

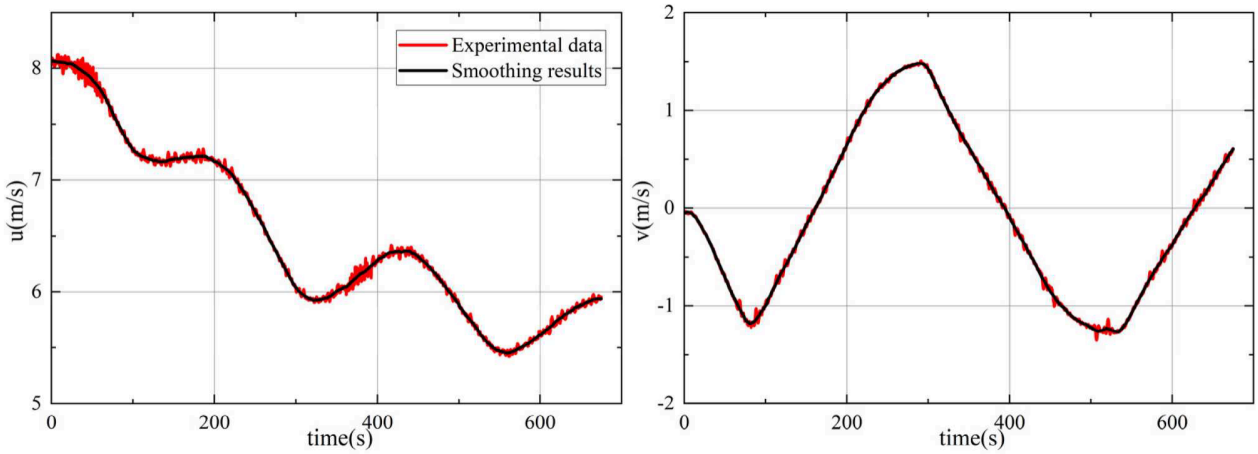


Figure 9. Smoothing results of the velocity values from the 20°/20°zigzag test (S). (This figure is available in color online.)

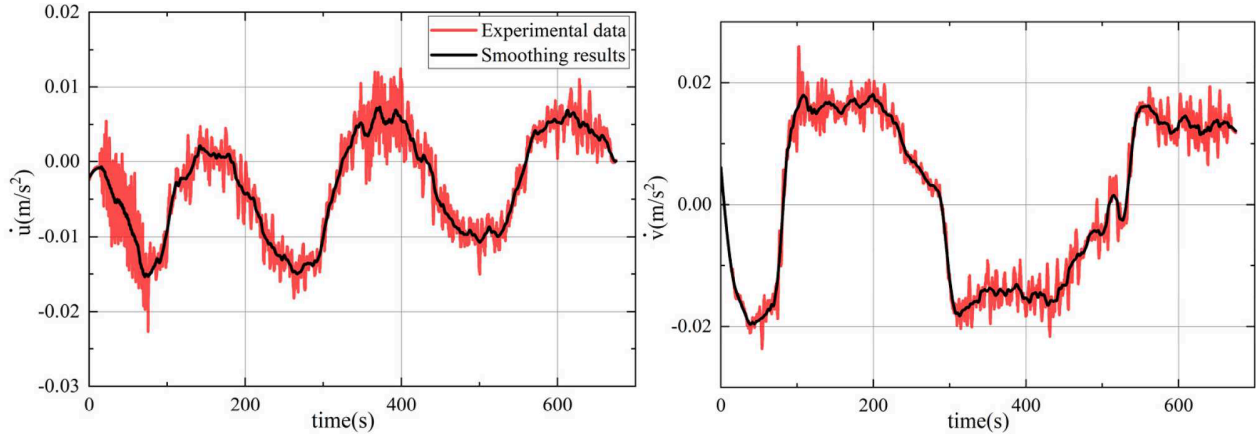
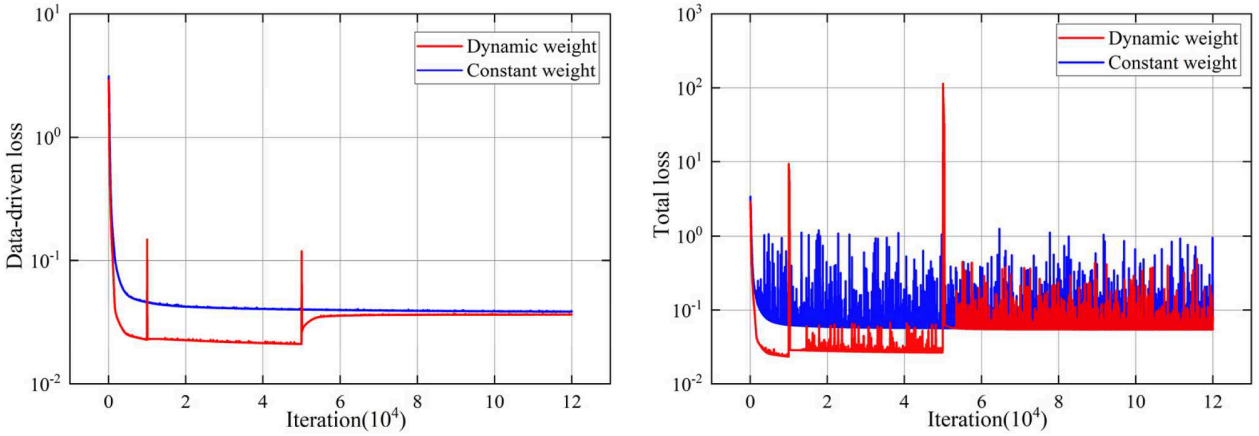


Figure 10. Smoothing results of the acceleration values from the 20°/20° zigzag test (S). (This figure is available in color online.)



(a) Convergence process of data-driven loss (b) Convergence process of total loss

Figure 11. Comparison of convergence process based on experimental data. (This figure is available in color online.)

Figure 11(b), the total loss with constant weight fluctuates more during the early stages, while the total loss with dynamic weight is more stable. At the end of training, the total loss with dynamic weight shows a generally lower value than that with constant weight. This suggests that the dynamic weight strategy not only stabilizes the training process but also improves the fitting accuracy.

Table 4. Identification of dimensionless hydrodynamic derivatives using PINN based on experimental data.

X-Coef.	Identified	Y-Coef.	Identified	N-Coef.	Identified
X'_u	−0.0029	Y'_v	−0.0197	N'_v	−0.0038
X'_{uu}	0.0013	Y'_{vvv}	−0.2613	N'_{vvv}	0.0230
X'_{vv}	0.0400	Y'_r	0.0057	N'_r	−0.0018
X'_{rr}	0.0104	Y'_{rrr}	0.0128	N'_{rrr}	−0.0020
X'_{vr}	0.0548	Y'_{vrr}	0.0547	N'_{vrr}	−0.0083
$X'_{\delta\delta}$	−0.0018	Y'_{vvr}	0.0322	N'_{vvr}	−0.0120
$X'_{v\delta}$	0.0075	Y'_δ	0.0044	N'_δ	−0.0012
$X'_{r\delta}$	0.0020	$Y'_{\delta\delta}$	0.0011	$N'_{\delta\delta}$	0.0001
		$Y'_{\delta\delta\delta}$	−0.0035	$N'_{\delta\delta\delta}$	−0.0011
		$Y'_{u\delta}$	−0.0026	$N'_{u\delta}$	0.0010
		$Y'_{v\delta\delta}$	−0.0099	$N'_{v\delta\delta}$	−0.0034
		$Y'_{vv\delta}$	0.0509	$N'_{vv\delta}$	−0.0007
		$Y'_{r\delta\delta}$	−0.0089	$N'_{r\delta\delta}$	−0.0031
		$Y'_{rr\delta}$	−0.0140	$N'_{rr\delta}$	−0.0017

In the prediction of 10°/10° zigzag test (S), as can be seen from Figure 12, the predictions from both PINN and identified models are in close agreement with the experimental data within the first 300 s. However, after 300 s, both models show a noticeable deviation. One possible reason is that the zigzag test data in this study is relatively limited, especially under small rudder angle conditions. This results in PINN failing to fully capture the ship's dynamic characteristics. As a result, the prediction error accumulates over time, leading to the observed deviation. Another possible reason is that the noise in the experimental data has not been fully removed. PINN may have overfitted to certain characteristics of the noise. In Figure 13, the predictions of both PINN and identified models for the 20°/20° zigzag

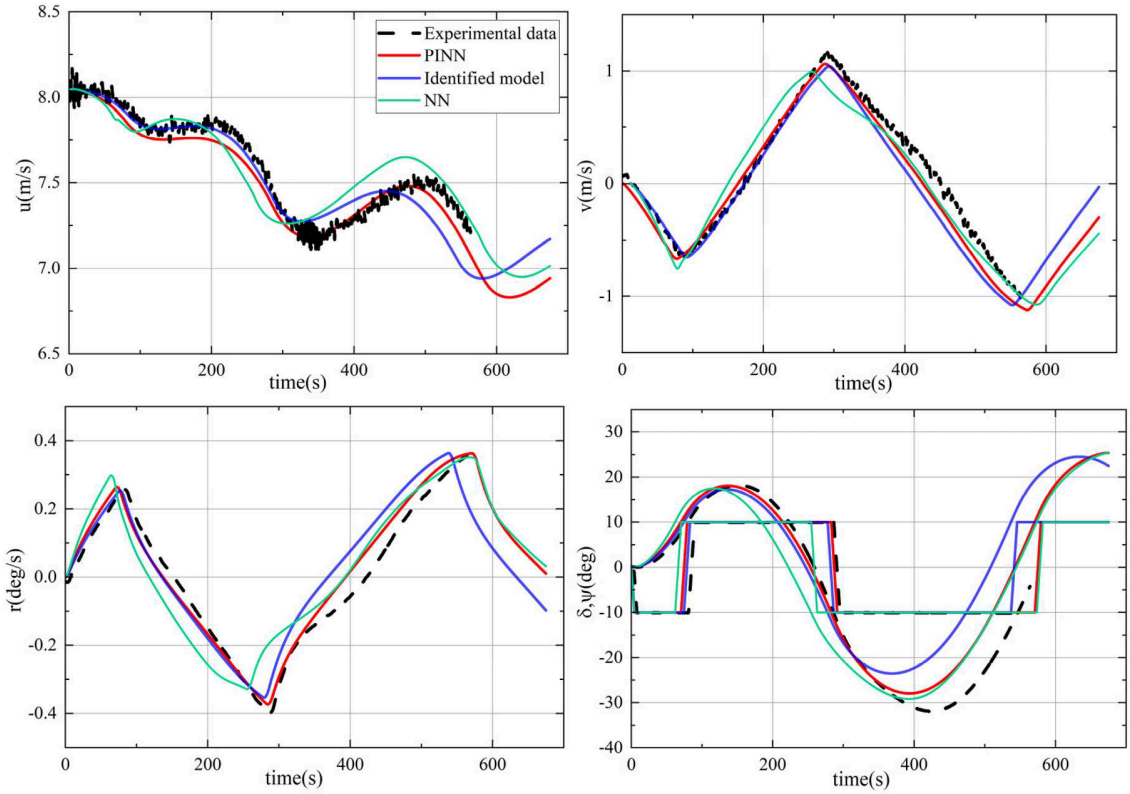


Figure 12. Prediction of 10°/10° zigzag test (S) based on experimental data. (This figure is available in color online.)

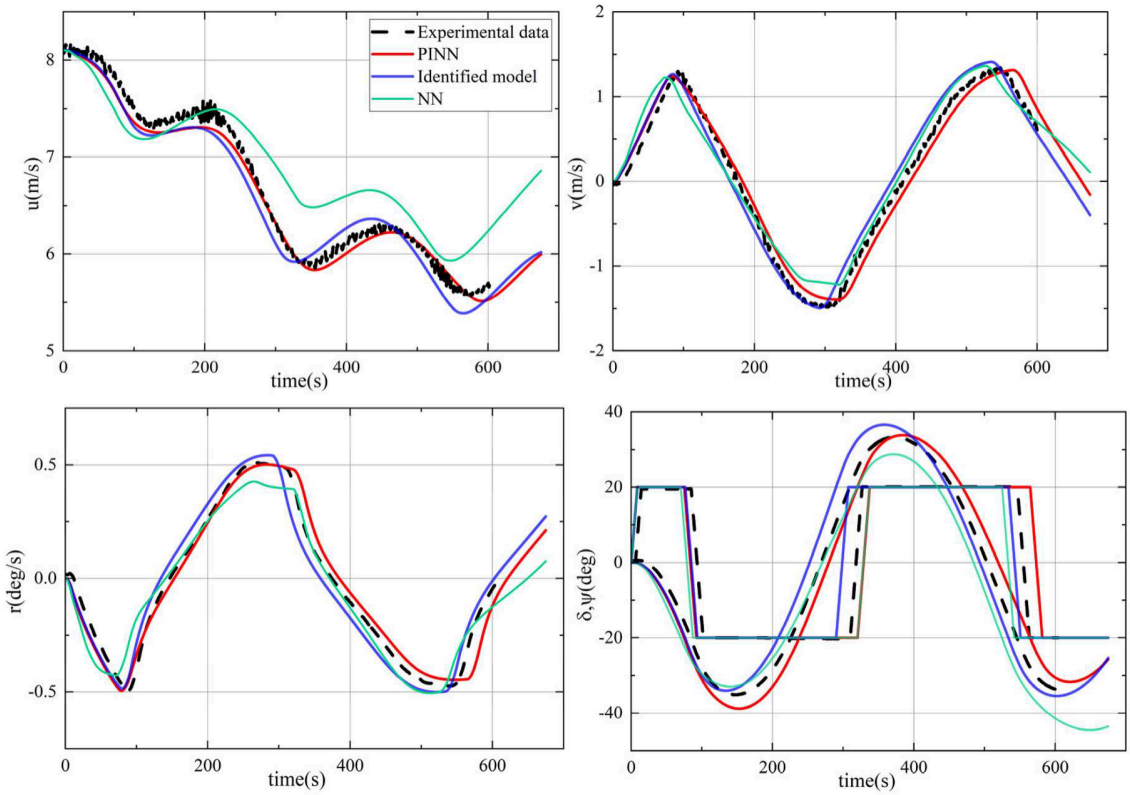


Figure 13. Prediction of 20°/20° zigzag test (P) based on experimental data. (This figure is available in color online.)

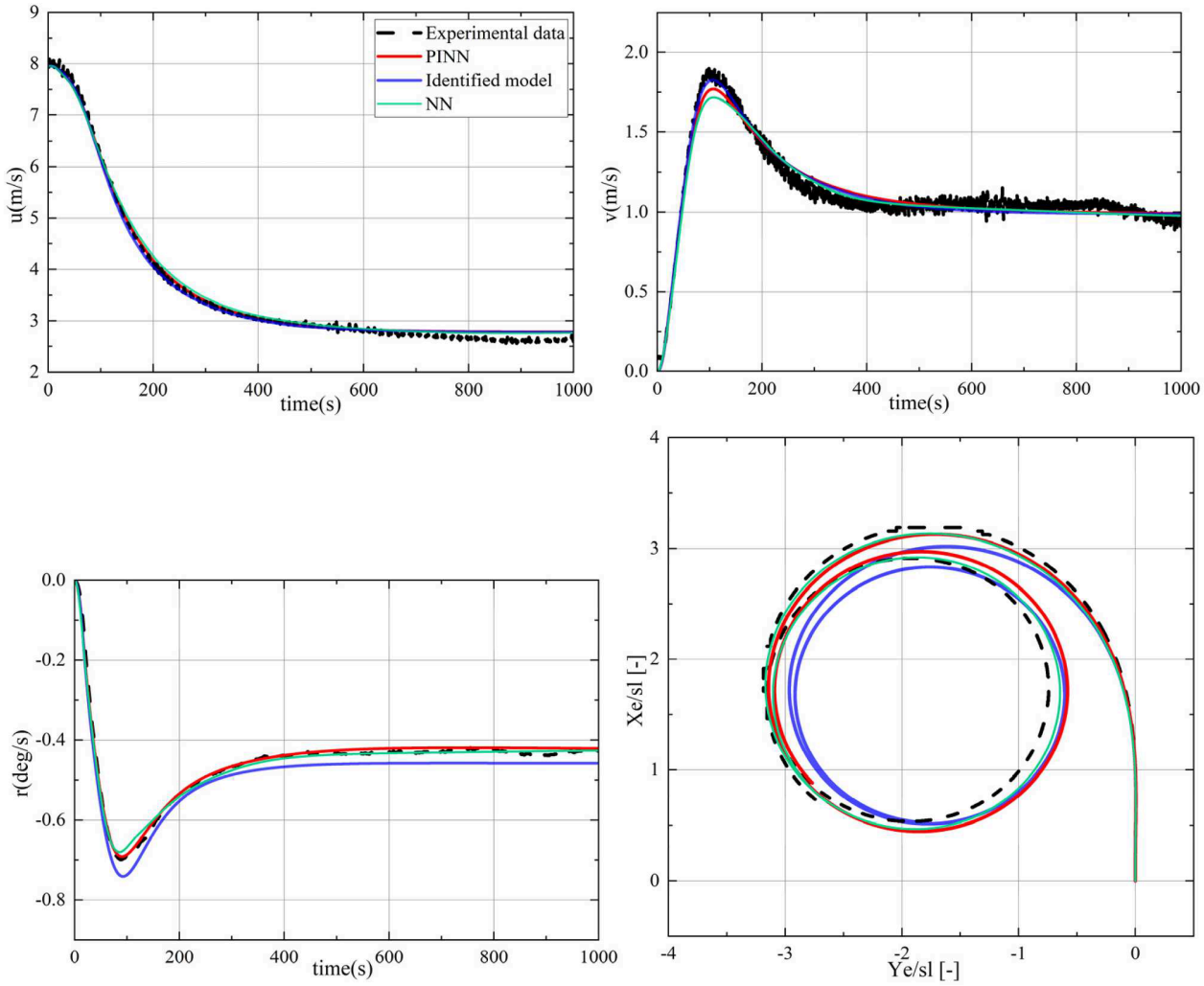


Figure 14. Prediction of 35° turning circle test (P) based on experimental data. (This figure is available in color online.)

test (P) exhibit only a small deviation. This may be due to the more pronounced dynamic changes at large rudder angles, which make it easier to capture the ship's dynamic characteristics. As shown in Figure 14, the prediction results for the turning circle test closely match the experimental data. This indicates that PINN captures the dynamic behavior at large rudder angles more accurately.

Table 5. Estimation of prediction accuracy by RMSE based on experimental data.

		u	*v*	*r*	sum
10°/10° zigzag (S)	PINN	0.0763	0.1272	0.0406	0.2441
	Identified	0.1075	0.2340	0.0735	0.4150
	NN	0.1311	0.2410	0.0866	0.4587
20°/20° zigzag (P)	PINN	0.1003	0.1349	0.0584	0.2936
	Identified	0.1579	0.1583	0.0682	0.3844
	NN	0.3617	0.1871	0.0756	0.6244
35° turning circle (P)	PINN	0.0907	0.0566	0.0085	0.1558
	Identified	0.0919	0.0539	0.0300	0.1758
	NN	0.0969	0.0568	0.0092	0.1629

According to the RMSE index in Table 5, PINN model shows better predictive accuracy than the NN model in all tests, while the identified model performs better in two tests and worse in one. On the one hand, this indicates that the generalization ability of the PINN model is superior to that of the traditional NN. For the NN, as shown in Figures 12–14, no significant deviation is observed in the NN predictions within the first 100 s. After this period, noticeable deviations gradually emerge. This behavior may be attributed to the accumulation and amplification of local approximation errors during iterative time integration. In contrast, physical constraints in the PINN suppress error accumulation, resulting in improved long-term forecasting accuracy. On the other hand, this reveals that the identification results of some hydrodynamic derivatives may still be inaccurate. Based on the sensitivity analysis in Section 4.3, although noise-free simulation data are used for training, some high-order

hydrodynamic derivatives are still prone to parameter drift. When training with experimental data containing noise, the parameter drift becomes even more severe (Wang et al. 2019). This leads to a reduced generalization ability of the identified model. In contrast, PINN is trained using the total loss from the physical equations, which helps reduce the impact of parameter shifts and enhances generalization performance.

Overall, although some hydrodynamic derivatives may exhibit identification errors, the PINN and the identified model perform better than the traditional NN model in most tests. This indicates that the inclusion of physical constraints has successfully improved the neural network's generalization performance, and the ship maneuvering motion modeling based on PINN is successful.

6. Conclusion

In this paper, a robust identification modeling method based on PINN is explored for ship maneuver prediction in calm water. Compared to nonparametric modeling, PINN improves the model's generalization performance by incorporating the ship dynamic model as a physical loss in the training framework. Concurrently, PINN can also reverse-solve for unknown hydrodynamic derivatives. To improve PINN performance, a dynamic weight coefficient is applied to the physical loss. The comparison with constant weight coefficient shows that the dynamic weight strategy is effective. Both the trained PINN model and the identified model are employed to predict the ship maneuvering motion.

To evaluate the effectiveness, two case studies are performed on the KVLCC2 ship: one using a dataset generated from simulations, and the other using the dataset of free-running model tests provided by the SIMMAN 2008 workshop. The experimental dataset is preprocessed by the Savitzky–Golay filter. The results from the simulation study indicate that predictions of both PINN and identified model match well with the original simulation data, exhibiting similar accuracy. Meanwhile, most identified hydrodynamic derivatives are close to the original values, while some higher-order derivatives deviate significantly. A sensitivity analysis is conducted, and the results show that these higher-order hydrodynamic derivatives are prone to parameter drift. The results from the experimental study show that both models perform better than the traditional neural network in most tests. They exhibit some deviation only at small rudder angles. This discrepancy may be due to the limited dataset for small rudder angle maneuvers. Additionally, the prediction accuracy of the identified model is lower than that of the PINN model. This suggests that the neural network with added physical constraints exhibits better generalization performance on noisy datasets. Overall, the PINN-based modeling method demonstrates satisfactory prediction accuracy.

Although the preliminary application of the PINN-based modeling method seems encouraging thus far, there are still some limitations. (1) The instability of PINN parameter identification under noise conditions. (2) Limited generalization ability caused by insufficient training data under the condition of small rudder angle. (3) The effect of Savitzky–Golay filtering on noise processing is limited, and there may still be some data noise interference. In addition, the PINN-based modeling method may be explored in the future for multi-condition scenarios. In these scenarios, ship maneuvering dynamics under various environmental factors – such as wind, waves, and current – may be incorporated, which could further enhance the model's robustness and applicability in real-world maritime conditions.

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Author contributions

CRediT: **Yi Hu:** Investigation, Software, Validation, Writing – original draft; **Shuo Liu:** Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing; **Tao Wang:** Methodology, Supervision, Writing – review & editing; **Yu Zhang:** Investigation; **Yong Cai:** Project administration.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The simulation data generated during this study have now been deposited in the Mendeley Data repository and are

available at the following DOI: https://doi.org/10.17632/nfrry82gsr.1.

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